Agent Based Modelling

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March 4, 2026

Flocking Model

Flocking model

- Birds move with constant velocity (v_0)
- Align themselves to neighbors
- Some noise due to inaccurate averaging
- Differential equation

$$\theta_i(t + \Delta t) = \langle \theta(t) \rangle_{|\mathrm{r}_i - \mathrm{r}_j| < R} + \xi$$

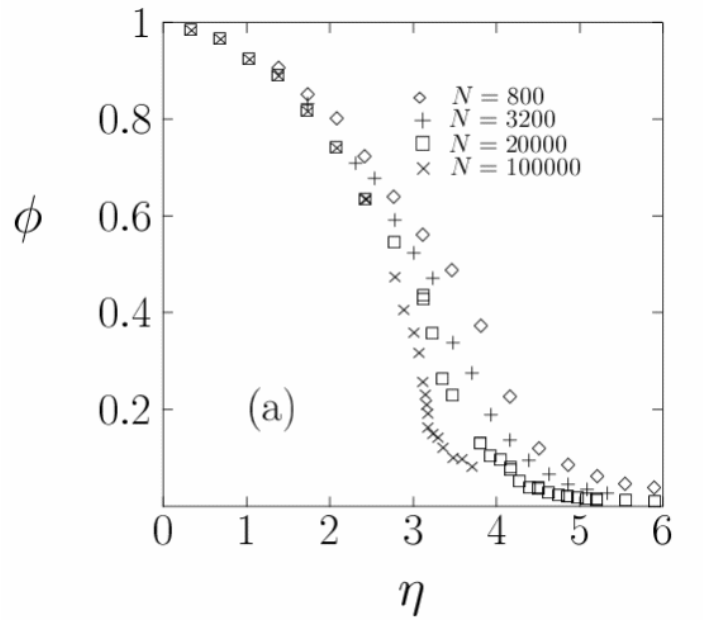
- Upgrade position:

$$\mathrm{r}_i(t + \Delta t) = \mathrm{r}_i(t) + v_0 \mathrm{e}(\theta_i(t)) \Delta t$$

where $\mathrm{e}(\theta)$ is a unit vector in the direction of θ.

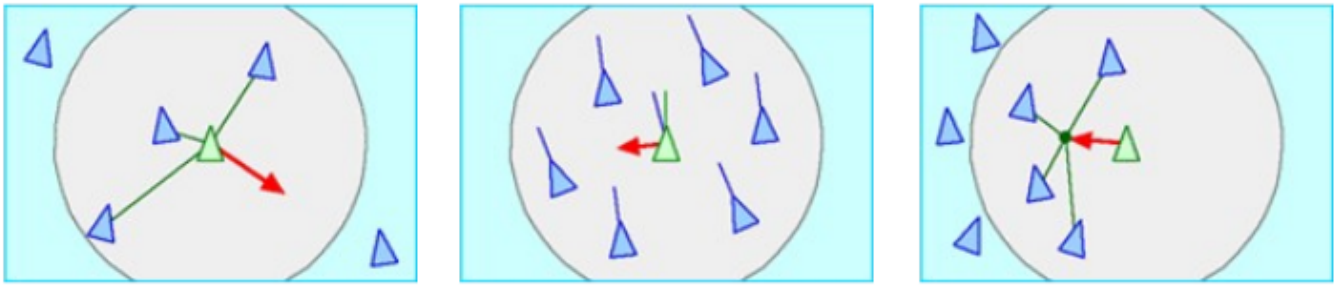
Flocking model

- Birds move with constant velocity (v_0)
- Align themselves to neighbors
- Some noise due to inaccurate averaging
- Phase diagram 1d:



Flocking model

- Birds move with constant velocity (v_0)
- Align themselves to neighbors
- Some noise due to inaccurate averaging
- Non-physicist model:



Separation: steer to avoid crowding local flockmates

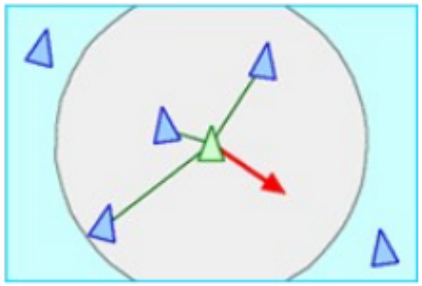
Alignment: steer towards the average heading of local flockmates

Cohesion: steer to move toward the average position of local flockmates

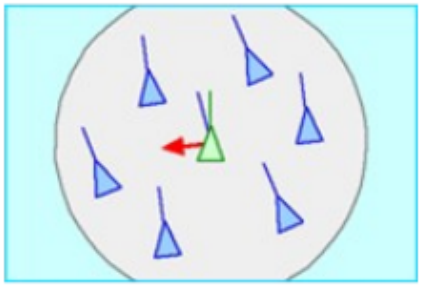
http://www.red3d.com/cwr/boids/

Practice

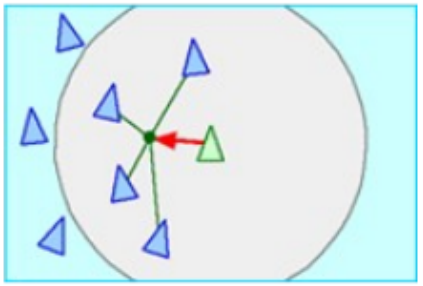
- Birds move with constant velocity (v_0)
- Align themselves to neighbors
- The also align themselves towards the center of their neighbor
- If they would get too close they turn away
- Velocity may vary from individual to individual
- There is a predator which has a larger coherence radius and only move towards the prey ahead of it
- Birds move away from the predator
- Birds are more inclined to align with a neighbor which did a big turn
- Some noise due to inaccurate averaging



Separation: steer to avoid crowding local flockmates



Alignment: steer towards the average heading of local flockmates



Cohesion: steer to move toward the average position of local flockmates

Videos

- https://www.youtube.com/watch?v=B2kQF7eWkIs
- https://www.youtube.com/watch?v=V4f_1_r80RY